

Supplementary Tables 6 and 7

Table 6 (manuscript Table 4) and Table 7 (manuscript Table 2). Bold values match the README layout.

Table 6 — Tri-stage restoration ablation and hyperparameter sensitivity

Restoration pipeline stage (Table 6)	Hyperparameters	mIoU (%)	mTC (%)
Raw input (no restoration)	—	65.4	77.2
+ Stage 1: flare suppression	$\tau_{\text{flare}} = 220$	67.3	79.5
+ Stage 1: flare suppression	$\tau_{\text{flare}} = 240$	68.6	81.0
+ Stage 1: flare suppression	$\tau_{\text{flare}} = 260$	67.9	79.4
+ Stage 2: DCP dehazing (guided)	$\sigma = 2.0$	70.2	83.2
+ Stage 2: DCP dehazing (guided)	$\sigma = 3.0$	71.4	84.6
+ Stage 2: DCP dehazing (guided)	$\sigma = 4.0$	70.6	83.0
+ Stage 3: CLAHE & gamma	$\gamma = 0.8$	71.2	85.9
+ Stage 3: CLAHE & gamma	$\gamma = 1.4$	73.1	86.8
+ Stage 3: CLAHE & gamma	$\gamma = 2.2$	71.6	86.2
Tri-stage photometric pipeline	—	74.8	88.4

Component-wise ablation and hyperparameter sweeps with SAM 3 + causal segmentation fixed (manuscript Table 4).

Table 7 — Comparative benchmark (excerpt)

Method (Table 7)	mIoU (%)	mTC (%)	FPS
SAM (per-frame zero-shot)	62.5	64.1	12.3
SAM 2 (streaming video)	71.6	85.1	22.4
Restoration + SAM 3 (causal PCS) — ours	74.8	88.4	28.6

State-of-the-art comparison on curated multi-scenario dashcam sequences (manuscript Table 2).